Lab Works

Required Materials:

1. 8085 Simulator or Kit

   https:;www.sim8085.com/

2. Programs

Lab1: Understanding addressing modes in 8085 microprocessors.

Topics: Direct addressing, Indirect addressing

1. **Write an ALP for 8085 MP to store an immediate 8-bit data in memory location 2051H using direct addressing.**
2. **Write an ALP for 8085 MP to store 8-bit data in memory 2501H using indirect addressing.**

Lab2: Addition with 8085 microprocessors.

1. **Write an ALP for adding two 8-bit numbers 02H and 03H using immediate addressing . The result is to be stored in 1050 H.**
2. **Write an ALP for adding two 8-bit numbers stored in 2501H and 2502H. The result is to be stored in 2503H.**
3. **Write an assembly language program to add two 8-bit numbers (F1 and 15), stored at C050H and C051H, with carry.**
4. **Write an assembly language program to add two 16-bit numbers A645 and 9B23, stored from C050H onwards, with carry.**

Lab3: Subtraction with 8085 microprocessors.

1. **Write an ALP for subtracting two 8-bit numbers stored in 2501H and 2502H. The result is to be stored in 2503H.**

Lab 4: Swapping of data with 8085 microprocessors.

1. **Write a program in Assembly language to interchange(Swap) the contents of two memory location 2100H and 2101H.**

Lab 5: Multiplication with 8085 microprocessors.

1. **Write a program in Assembly language to multiply two 8 bit numbers stored at C050H and C051H.**

Lab 6: Looping in 8085 programs

1. **Write a program in Assembly language to find the square of a number stored at 8000H.**
2. **Write a program in Assembly language to print N natural numbers upto the number stored at 8000H. (ie. If 5, then the series should be 1,2,3,4,5)**
3. **Write a program in Assembly language to find the factorial of a number stored at 8000H and store the factorial at 8001H.**

5. Basic Assembly language programs of 8085

Test Ok

1. Write an ALP for 8085 MP to store an immediate 8-bit data in memory location 2051H using direct addressing.

```
MVI A,49H      ; "Store 49H in the accumulator"
STA 2501H      ;"Copy accumulator contents at address 2501H"
HLT            ; "Stop"
```

Test
Ok

2. Write an ALP for 8085 MP to store 8-bit data in memory 2501H using indirect addressing.

```
LXI H,2501H        ; "Load H-L pair with 2501H"
MVI M,49H          ;  "Store 49H in memory location pointed by H-L register pair (2501H)"
HLT                ;  "Stop"
```

Test Ok

3. Write an ALP for adding two 8-bit numbers 02H and 03H using immediate addressing . The result is to be stored in 1050 H.

```
MVI A, 02H      ; “Get the first number and move to accumulator”
MVI B, 03H      ;  “Get the second number and move to register B"
ADD B           ; "Add the second number with the content of accumulator"
STA 1050H       ;  "Store result at 1050H"
HLT             ;  "Stop"
```

4. Write an ALP for adding two 8-bit numbers stored in 2501H and 2502H. The result is to be stored in 2503H.

Test Ok

```
LXI H, 2501H   ; "Get address of first number in H-L pair. Now H-L points to 2501H"
MOV A, M       ; "Get first operand in accumulator"
INX H          ; "Increment content of H-L pair. Now, H-L points 2502H"
ADD M          ;  "Add first and second operand"
INX H          ;  "H-L points 2503H"
MOV M, A       ;  "Store result at 2503H"
HLT            ; "Stop"
```

Test Ok

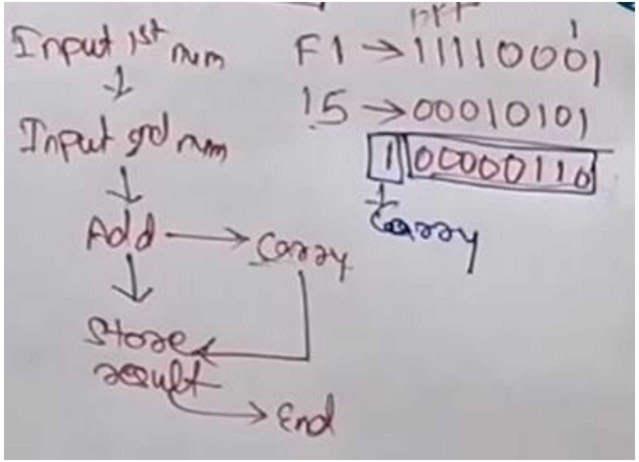
5. Write an ALP for subtracting two 8-bit numbers stored in 2501H and 2502H. The result is to be stored in 2503H.

```
LXI H, 2501H   ;  "Get address of first number in H-L pair. Now H-L points to 2501H"
MOV A, M       ;  "Get first operand in accumulator"
INX H          ;  "Increment content of H-L pair. Now, H-L points 2502H"
SUB M          ;  "Subtract first to second operand"
INX H          ;  "H-L points 2503H"
MOV M, A       ;  "Store result at 2503H"
HLT            ;  "Stop"
```

Test Ok

6. Write an assembly language program to add two 8-bit numbers (F1 and 15), stored at C050H and C051H, with carry.

```
        LXI H, C050H  ; "Get address of first number in H-L pair. Now H-L points to C050H"
        MOV A, M      ;  "Get first operand in accumulator from C050H"
        INX H         ;  "Increment content of H-L pair. Now, H-L points C051H"
        MVI B, 00H    ; "Initialize Register B with 00H for carry store"
        ADD M         ;  "Add first and second operand, result stored in Acc"
        JNC label     ; "Jump to label if no carry"
        INR B         ; "Increment content of B by 1"
label:  STA C053H     ; "Store the content of Acc to C053H"
        MOV  A, B     ;  "Move carry at Acc"
        STA C052H     ; "Store the content of Acc to C052H, ie the carry part"
        HLT           ;  "Stop"
                      ; "Carry part store in C052H and sum is store in C053H"
```

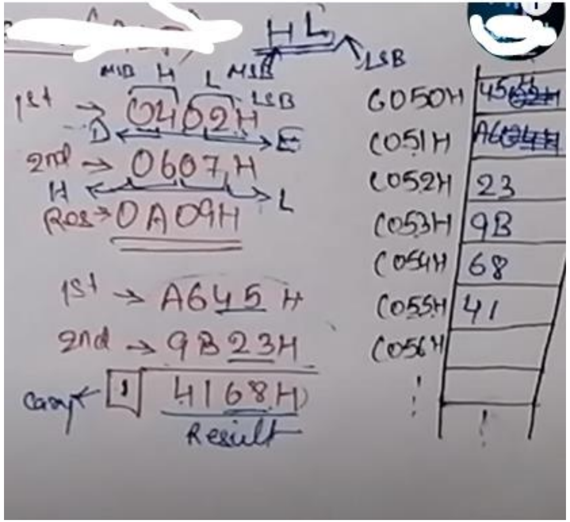


Reference:
https:;www.youtube.com/watch?v=GLfEQGY70Ek&list=PLndX8heiWwErijxt7FpnRilO2KV4UhFmN

7. Write an assembly language program to add two 16-bit numbers A645 and 9B23, stored from C050H onwards, with carry.

Test Ok

```
        LHLD C050H   ; "Get address of first number from memory and load in in H-L register"
        XCHG         ; Exchange H and L with D and E
        LHLD C052H   ;  "Get address of second number from memory and load in in H-L register"
        MVI B, 00H   ; "Initialize Register B with 00H for carry store"
        DAD D        ; Add both the register pairs and store in HL, ie. HL+DE
        JNC label    ; "Jump to label if no carry"
        INR B        ; "Increment content of B by 1"
label:  SHLD C054H   ; "Store the content of H-L register to C054H"
        MOV  A, B    ;  "Move carry at Acc"
        STA C056H    ; "Store the content of Acc to C056H, ie the carry part"
        HLT          ;  "Stop"
                     ; "Carry part store in C056H and sum is store in C054H-C055H"
```



Reference: https:;www.youtube.com/watch?v=_4Q-upoIv8c&list=PLndX8heiWwErijxt7FpnRilO2KV4UhFmN&index=2

Test Ok

8. Write a program in Assembly language to interchange(Swap) the contents of two memory location 2100H and 2101H.

```
LDA 2100H     ; “Load the content of 2100H into Accumulator”
MOV B,A       ; “Move content of Acc to register B”
LDA 2101H     ;“Load content of 2101H to Accumulator”
STA 2100H     ; “Store content of Acc. to 2100H”
MOV A,B       ; “Move content of register B to Acc.”
STA 2101H     ; “Store content of Acc. To 2101H”
HLT           ; “End”
```

Test
Ok

8. Write a program in Assembly language to multiply two 8 bit numbers stored at C050H and C051H.

```
        LXI H,C050H     ; Get address of first number in H-L pair. Now H-L points to C050H"
        MOV B,M         ; Get first operand in register B from C050H"
        MVI A,00H       ;Clear Accumulator
        MVI D, 00H      ;Initialize a register D with 00H for carry storage
        INX H           ; point to C051H
        MOV C,M         ;Get the second operand
loop1:  ADD B           ;Add contents in register B with Accumulator
        JNC loop2       ; Jump to loop2 if no carry
        INR D           ;increase D
loop2:  DCR C           ;Decrement counter C
        JNZ loop1       ;Jump to loop1 if C not equal to 0
        STA C052H       ;Store the content to C052H
        MOV A,D         ;Move D to A.
        STA C053H       ;Store content of Acc. To C053.
        HLT
```

5x3*15

5+5+5=15

Test Ok

9. Write a program in Assembly language to find the square of a number stored at 8000H.

```
        LXI H,8000H     ;Load the number from 8000H
        XRA A           ;Clear accumulator
        MOV B,M         ;Load data from memory to B
LOOP:   ADD M           ;Add memory byte with A
        DCR B           ;Decrease B by 1
        JNZ LOOP        ;jump to loop if B not equal to 0
        STA 8050H       ;Store result into memory
        HLT             ; Terminate the program
```

Note: This finding of square of a number n is done by adding the numbers for n times in the accumulator.

Test Ok

10. Write a program in Assembly language to print N natural numbers upto the number stored at 8000H. (ie. If 5, then the series should be 1,2,3,4,5)

```
        LXI H, 8000H    ;Load the number store at 8000H
        MOV B,M         ;Take number from memory to B
        MVI D, 01H      ;Set D to 01H
loop:   ADD D           ;Add content in D to Accumulator
        INX H           ;Increase the pointer to next memory location
        MOV M,A         ;Move content of Accumulator to memory
        DCR B           ;Decrease B by 1
        JNZ loop        ;Jump to loop if B not equal to 0
        HLT             ;Terminate the program
```

Test
Ok

11. Write a program in Assembly language to find the factorial of a number stored at 8000H and store the factorial at 8001H.

```
        ;Main program
        LXI H, 8000H        ;Load the number store at 8000H
        MOV B,M             ;Take number from memory to B
        MVI D, 01H  ;Set D to 01H
FACT:   CALL MUL            ;Call multiplication subroutine
        DCR B               ;Decrease B by 1
        JNZ FACT            ;If Z = 0, jump to Fact
        INX H               ;Point to next location
        MOV M,D             ;Store result in memory
        HLT                 ;Terminate the program

        ;Sub Program1
        MUL:    MOV E,B     ;Load B to E
                XRA A       ;Clear accumulator to store result

        ;Sub Program2
        ML:     ADD D       ;Add D to A
                DCR E       ;Decrement E
                JNZ ML      ;If Z = 0, jump to ML
                MOV D,A     ;Transfer contents of A to D
                RET         ;Return result
```

Note: This finding of factorial of a number n is done by successive addition using nested looping.

The End